

Graph Neural Networks and Graph Foundation Models for Scientific Discovery

Abstract

Scientific discovery is intrinsically relational. Molecules are atoms connected by bonds and spatial interactions, crystals are periodic atomistic neighborhoods, proteins are contact- and geometry-constrained residue systems, cells communicate through regulatory and spatial graphs, and many physical simulators operate on interaction graphs or meshes. Graph neural networks (GNNs) therefore occupy a distinctive place in AI for science: they are not merely convenient predictors for irregular data, but a family of models whose inductive biases align closely with scientific structure. Over the past decade, the field has moved from spectral graph convolutions and message-passing neural networks to equivariant geometric architectures, graph transformers, multimodal graph-language models, and increasingly large pretraining regimes. In parallel, the idea of graph foundation models (GFMs) has emerged to denote graph-native or graph-enhanced models pretrained at scale and adapted across tasks, datasets, and sometimes modalities. Existing evidence from chemistry, materials science, structural biology, spatial omics, and physical simulation shows that this trajectory is already accelerating real scientific workflows, from property prediction and docking to atomistic simulation, protein design, and weather forecasting [1-48,49-115,140-165,196-201]. ¹

This survey synthesizes that literature through a single question: how have graph models evolved from domain-specific predictive tools into increasingly general-purpose scientific infrastructures? The answer, developed across more than 200 references, is that GNNs and GFMs are converging around four themes: stronger geometric and physical inductive bias; larger and more diverse pretraining corpora; multimodal grounding in text, sequences, knowledge graphs, and experimental context; and adaptation mechanisms aimed at data-scarce, distribution-shifted, and high-stakes scientific regimes. At the same time, the survey argues that most current “foundation” claims remain domain-bounded: many are large, reusable pretrained graph encoders rather than fully universal graph models. Scientific progress will therefore depend not only on scaling, but also on trustworthy evaluation, uncertainty quantification, causal and mechanistic grounding, interoperable multimodal representations, and integration with closed-loop experimentation [25-48,77-83,103-115,152-201]. ²

Introduction

Graph learning became central to modern AI because many real systems are neither sequences nor regular grids. In science this observation is stronger: graphs are often the natural formal language of the problem itself. Atomistic systems are graphs over atoms and bonds or over atoms and local geometric neighborhoods; protein structures can be represented as residue graphs, atom graphs, surface graphs, and contact maps; cellular and biomedical systems are naturally described by interaction, regulation, communication, and spatial graphs; and physical simulation on meshes or particle systems is graph-structured by construction [1-15,49-76,84-115,140-150]. ³

The first phase of graph deep learning for science was dominated by supervised, task-specific GNNs. Message passing proved especially effective because local relational aggregation corresponds naturally to chemical neighborhoods, coordination shells in materials, and residue-level interaction patterns in proteins. Early and mid-generation systems such as neural message passing for quantum chemistry, SchNet, CGCNN, MEGNet, DimeNet, and related architectures established that graph-based inductive bias could outperform handcrafted descriptors while retaining a plausible degree of structural interpretability [7,16-24,49-76]. In chemistry and materials, this development was so influential that recent reviews explicitly identify GNNs as a major organizing paradigm of machine learning in those fields [61,62]. 4

The second phase introduced geometry, symmetry, and physics more directly into the model class. Three-dimensional molecular prediction, interatomic potentials, protein docking, and atomistic simulation all exposed limitations of purely topological GNNs. This led to directional message passing, invariant and equivariant neural networks, graph transformers with geometric encodings, and models that predict forces, energies, conformations, or binding poses while respecting Euclidean symmetries [24,58-77,84-91,116-145,187]. These models drove some of the most visible AI-for-science advances of the early 2020s, including universal or near-universal interatomic potentials, protein structure prediction and inverse folding, sequence design, and graph-based weather forecasting [58,59,63,92-100,109,127,145]. 5

The current phase is characterized by scale and transfer. Inspired by foundation models in language and vision, graph learning has moved toward large-scale self-supervision, graph masking, graph-text alignment, multimodal pretraining, and cross-dataset adaptation. In the graph community, this shift is now explicitly framed as graph foundation modeling: the ambition is to pretrain models on extensive graph corpora and then adapt them across tasks, datasets, and domains [41-48,159-165,198]. In scientific discovery, this ambition is visible in molecular and materials pretraining, multimodal molecule-text models, reusable atomistic backbones, spatial-omics graph transformers, and attempts at more universal graph-structured biomedical models [30,36-48,64,77-83,103-115,154-165,196-201]. Yet the field is heterogeneous. Some models are clearly graph-native; others are graph-language hybrids; still others are domain-specific foundation models whose “graphness” is one modality among several. This survey treats all of these as part of a single scientific transition, while distinguishing between genuine cross-task reuse and narrower domain-specific transfer. 6

Against this backdrop, the purpose of this survey is threefold. First, it revisits the methodological trajectory from classical GNNs to contemporary GFMs, emphasizing which architectural ideas matter for scientific data rather than for generic benchmark performance. Second, it organizes the literature into coherent research directions spanning representation learning, pretraining, multimodality, generation, and scientific deployment. Third, it identifies the practical and conceptual bottlenecks that separate current large pretrained graph models from the more ambitious vision of scientific foundation models that are robust, mechanistically meaningful, and genuinely reusable across discovery workflows [13-15,41-48,104,151,159-165,196-201]. 7

Methods

A useful way to understand this literature is to begin with **graph construction and scientific inductive bias**. Scientific graphs are not interchangeable. In small-molecule learning, one may use 2D bond graphs, 3D conformer graphs, or multimodal combinations of sequence-like chemical strings, graph topology, and geometry. In materials science, crystal graphs and line graphs must encode periodicity, bond angles, lattice information, or long-range electrostatics. In structural biology, residue graphs, atom graphs, surface

graphs, and interaction graphs each expose different biological priors. In spatial omics, graphs are often hierarchical: a tissue can be a cell-neighborhood graph while each cell contains a gene-regulatory or co-expression graph. These design choices are often more consequential than modest changes in downstream optimizer or backbone hyperparameters because they determine what invariances, locality assumptions, and sources of scientific information the model can access [49-76,84-115,140-150,167-174]. 8

The backbone families used on these graphs can be grouped into **message-passing networks, graph transformers, and geometric/equivariant models**. Message-passing neural networks remain the canonical scientific GNN because local aggregation maps naturally onto bonded neighborhoods and interaction shells [7,49-57,65]. In chemistry and materials, the progression from SchNet and CGCNN to DimeNet, GemNet, M3GNet, CHGNet, and related models reflects a steady enrichment of geometric information: continuous filters, directional information, spherical harmonics or analogous angular structure, and in many cases force-consistent or energy-based outputs [54-76]. In biology, comparable logic underlies graph models for function prediction, docking, inverse folding, and interface prediction [84-96]. Graph transformers broaden the receptive field and can alleviate oversquashing and long-range dependency limitations, while sparse or hybrid designs try to retain scalability [20-23,58-77,84-91,140-145,187]. Equivariant architectures add a still stronger scientific prior by ensuring that predictions transform correctly under rotations and translations, which is essential when the target depends on 3D structure rather than on graph topology alone [24,58,64,70,73-75,84-91,124-131,187]. 9

A second family centers on **self-supervised learning and graph pretraining**. In non-scientific graphs, graph contrastive learning, masking, and masked autoencoding were motivated by label scarcity. In scientific discovery, the motivation is even stronger: labels are often expensive, noisy, assay-dependent, or domain-shifted, whereas unlabeled structures are abundant. This is why molecular and atomistic graph learning became one of the earliest areas where graph pretraining achieved clear practical relevance. Strategies for pre-training graph neural networks, GraphCL, GCC, GROVER, GraphMAE, GraphMAE2, MoleBERT, GraphMVP, and KPGT form the core methodological lineage [25-40]. The scientific variant of this story is that pretraining objectives are rarely purely generic. They often exploit chemical motifs, 2D-3D agreement, local geometry, substructure masking, or domain knowledge encoded through curated tasks and auxiliary signals [30,36-39,64,77-83,123,136-139,201]. 10

From this line of work emerged **graph foundation models proper**, which seek broader transfer across tasks and datasets. In the general graph literature, recent models such as Graph Foundation Models Are Already Here, One for All, GFT, ZeroG, and cross-domain graph-pretraining methods formulate transfer as a first-class design objective [41-48,198]. Their mechanisms include task-unified prompting, transferable structural vocabularies, feature alignment across graphs, graph tokenization, and generative pretraining framed analogously to language modeling [41-48,164,165]. For scientific discovery, however, the most influential GFM are often **domain-specific** rather than universal. Uni-Mol, MoleculeSTM, MolFM, molecule-text aligned models, large atomistic pretraining backbones, HEIST, and emerging biomedical GFMs show that transfer becomes substantially easier when models can reuse chemically or biologically coherent vocabularies and modalities [64,77-83,97-100,103-115,156,196-201]. This suggests that, at least in the medium term, the most scientifically useful GFMs may be “meso-scale” foundation models: broad enough to transfer across many tasks within a scientific regime, but not yet universal across all graph domains. 11

A third family comprises **multimodal graph-science models**. Scientific reasoning rarely depends on structure alone. Molecules come with textual descriptions, assay metadata, spectra, patents, and biomedical context; proteins come with sequences, structures, mutational landscapes, and literature;

materials are connected to synthesis narratives, crystal text descriptions, and simulation/experimental outputs. Consequently, many recent scientific GFM are not graph-only but graph-plus-language, graph-plus-sequence, or graph-plus-knowledge-graph systems [78-83,92-100,113-115,154-165,196-201]. MoleculeSTM aligns structure with text; MolFM incorporates text and knowledge-graph context; UniMoT treats molecule-text interaction more like autoregressive language modeling; and HEIST learns across tissue graphs and gene-regulatory graphs [78-83,114]. These systems matter because they broaden the types of supervision available to scientific models and move them closer to scientific use in retrieval, editing, question answering, reaction prediction, and hypothesis generation rather than restricting them to scalar property prediction alone [78-83,113-115,156,159-165,197-201]. ¹²

A fourth family focuses on **generation, inverse design, and closed-loop discovery**. Here, graphs cease to be only inputs and become the space in which hypotheses are proposed. Early molecular generation methods such as junction-tree VAEs, MolGAN, GraphAF, and graph-to-graph translation methods established the feasibility of generating valid chemical structures or optimizing existing molecules [116-122,189,193-195]. Diffusion models then made graph-based or geometry-aware generation markedly more competitive for conformers, docking poses, linkers, and proteins [124-131]. In structural biology, inverse folding, docking, and sequence design are now tightly connected to graph/geometric learning [90-96,109,127,128,132,133]. In materials and chemistry, these generative models increasingly appear as components in larger pipeline systems that include retrieval, language guidance, physical filtering, and sometimes autonomous or semi-autonomous experimentation [63,64,77-83,116-139,196-201]. The key methodological shift is that scientific GNNs and GFMs are no longer just discriminative or predictive; they are becoming proposal engines for scientific search in structured design spaces. ¹³

These method families are most visible in **chemistry and drug discovery**, where graph learning has arguably matured furthest. Graph models support molecular property prediction, drug-target affinity, molecular optimization, quantum-chemical surrogate modeling, docking, reaction prediction, and de novo design [49-53,65-83,90,95,106-108,116-139,172-176,191-195]. The chemistry literature also helped pioneer graph pretraining at scale through GROVER, GraphMVP, KPGT, Mole-BERT, Uni-Mol, and multimodal molecule-text models [30,36-39,77-83]. Importantly, this domain illustrates both the strength and weakness of graph learning. Strength arises from close alignment between chemical structure and graph representation. Weakness arises because many biologically central endpoints depend on assay context, conformational ensembles, solvent, protein environment, and measurement protocol, which no single graph can fully capture. Thus current molecular GFMs are best understood as **cross-modal scientific priors** rather than exhaustive world models of chemistry [53,77-83,106-108,172-176]. ¹⁴

In **materials science**, GNNs became decisive because crystals and atomistic simulations demand learned representations that preserve local coordination and geometric interaction. CGCNN, MEGNet, ALIGNN, M3GNet, CHGNet, GemNet-OC, and related architectures enabled property prediction, relaxation, force prediction, and universal or broad interatomic potentials [54-76]. Large-scale training pushed this trajectory further. Merchant et al. showed that scaled graph networks can materially improve discovery efficiency, while recent materials-science surveys position foundation models, LLM agents, and graph-based atomistic models as a common infrastructure for design, simulation, and process planning [63,196]. The materials case is especially important for graph foundation modeling because it provides one of the clearest examples where a pretrained graph-based model becomes a reusable simulator-like backbone rather than merely a benchmark classifier [58,59,63,64,167-170,186,189,190,196,198-200]. ¹⁵

In **biology and biomedicine**, graphs appear at multiple scales: molecular, structural, cellular, patient-level, and knowledge-graph level. DeepFRI, EquiBind, ProteinMPNN, AlphaFold-adjacent geometric systems, scGNN, PTGB, HEIST, and biomedical graph-representation learning studies show that graph structure can organize diverse biological tasks from functional annotation to spatial tissue analysis [84-115,156-158]. Yet biology also exposes a major methodological boundary. Unlike atoms in a molecule, biological nodes and edges are often assay-defined, context-specific, and institution-specific. This is why HEIST’s hierarchical graphs and the recent attempt toward a universal graph-structured biomedical foundation model are especially notable: they are attempts to **factor out dataset-specific features** and retain only more transferable relational signatures [114,115]. If successful, this may become one of the most important directions for scientific GFMs because biology is precisely where data heterogeneity is greatest and where generic transfer is hardest [101-115,156-158]. ¹⁶

Finally, in **physics, environmental science, and simulation**, graphs function as learned discretizations of interacting systems. Interaction Networks, graph-network simulators, MeshGraphNets, neural PDE solvers, and graph-based weather forecasting treat the graph as a dynamic computation substrate whose edges represent physical coupling rather than semantic relation [140-150]. This literature is indispensable for the future of scientific GFMs because it highlights two underemphasized facts. First, scientific functionality often requires long-horizon stability, conservation-aware behavior, and out-of-distribution extrapolation, not only one-step prediction accuracy. Second, graph models can act as differentiable simulators or surrogate solvers, moving them closer to the notion of a scientific engine rather than a classifier [140-150]. The success of GraphCast underscores the point: graph architectures can already compete in real scientific forecasting pipelines when the representation and training objective are well aligned with the scientific process [145]. ¹⁷

Challenges and Prospects

The central challenge for both GNNs and GFMs in science is **heterogeneity**. Scientific graphs vary in node semantics, edge meaning, scale, sparsity, dimensionality, and annotation regime. A molecule graph, a catalyst surface graph, a protein-contact graph, and a cell-neighborhood graph all share relational form but differ profoundly in what a node or edge means. This makes naive cross-domain transfer brittle. Recent GFM work responds with structural tokenization, transferable tree vocabularies, task-unified formulations, and feature-agnostic prompts [41-48,114,115,154-165,198]. The broader lesson is that successful graph foundation modeling will likely require **representation interfaces** that separate transferable structure from domain-specific metadata, rather than assuming that more pretraining alone will solve cross-domain incompatibility. ¹⁸

A second challenge is **evaluation realism**. Scientific graph models are still too often judged on static benchmark splits that understate temporal drift, assay shift, scaffold shift, experimental confounding, or deployment mismatch. MoleculeNet, OGB, the Open Catalyst datasets, TDC, DrugOOD, GOOD, and newer materials discovery benchmarks have improved rigor, but they also make clear how sensitive outcomes are to dataset construction, splitting strategies, and proxy-task choice [166-178,190]. Models that perform well on random or even scaffold splits may still fail on the kinds of extrapolation that scientific discovery actually requires: new chemistry classes, new crystal prototypes, new tissues, new laboratories, or new forecasting regimes. If graph foundation models are to justify the “foundation” label in science, they must demonstrate robust cross-distribution transfer under evaluation protocols that reflect scientific novelty rather than benchmark familiarity [44-48,166-178,190,196-200]. ¹⁹

A third challenge is **trustworthiness**, including uncertainty, interpretability, and mechanistic plausibility. High-stakes scientific settings cannot rely solely on point estimates. Drug design, catalyst discovery, medical decision support, and autonomous experimentation require confidence estimates, failure detection, and interpretable rationales [179-186]. GNN explanations such as GNNExplainer, PGExplainer, and CF-GNNExplainer have advanced subgraph-level interpretability, but scientific use demands more than post hoc explanations: it demands physical consistency, calibrated uncertainty, and explanations that map onto known mechanisms or experimentally testable hypotheses [179-186]. This is especially urgent for foundation-scale models, where scale and multimodality can increase performance while simultaneously obscuring causal or mechanistic accountability. ²⁰

A fourth challenge is **scaling without losing scientific structure**. The language-model analogy is useful but incomplete. In science, larger models are not always better if they dilute symmetries, conservation laws, or domain semantics. The most successful scaled graph models in science have typically combined scale with strong structure: large graph networks for materials discovery, charge-informed interatomic potentials, large 3D molecular pretraining, and graph-based weather models all preserve scientific geometry rather than replacing it with generic sequence modeling [58,59,63,64,77-83,145,177,196-201]. The prospect here is not that graph models become linguistically generic, but that they become **structurally general**: reusable across many tasks because they faithfully encode the geometry and relations that science repeatedly reuses. ²¹

The most promising future direction is therefore **closed-loop, multimodal, scientifically grounded graph foundation modeling**. In such systems, graph encoders or graph-language hybrids would not operate in isolation. They would interact with literature retrieval, sequence and structure models, simulators, causal or symbolic modules, experiment planners, and uncertainty-aware decision rules. This is already visible in nascent form in molecular multimodal models, materials foundation-model discussions, emerging graph-language reasoning systems, spatial-omics GFMs, and autonomous materials-discovery frameworks [78-83,114,115,154-165,196-201]. The long-term opportunity is to make graph models the relational backbone of AI scientists: the component that keeps track of entities, interactions, constraints, and design spaces while other modalities contribute semantics, narration, and planning. The field's immediate task is to make that backbone reliable enough for genuine scientific use. ²²

Conclusion

The literature reviewed here shows that graph neural networks have already become one of the core methodological substrates of AI for science. Their appeal is not accidental: graph induction matches the relational fabric of scientific objects, from molecular structure and crystalline periodicity to biomolecular interfaces, tissue architecture, and interacting physical systems. This structural fit explains why graph methods have had durable impact across chemistry, materials, biology, and simulation, and why many of the most influential recent AI-for-science systems are graph-centric or graph-enhanced [1-24,49-115,140-150]. ²³

At the same time, the transition from powerful scientific GNNs to genuine graph foundation models remains incomplete. The strongest current systems are often domain-specific, multimodal, and highly reusable, but not yet universal. That is not a weakness so much as a realistic diagnosis of where the field stands. Scientific discovery imposes stricter requirements than consumer AI: models must generalize under deeper distribution shift, operate with calibrated uncertainty, respect domain symmetries and mechanisms, and support real experimental decision making. The future of graph foundation modeling will therefore be

defined less by abstract scale than by whether graph-based pretrained systems become trustworthy scientific infrastructures for transfer, reasoning, and closed-loop discovery [41-48,77-83,103-115,154-165,196-201]. ²⁴

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